

# Chapter 5: The Real Numbers

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## A: Cauchy Sequences

### Definition 1 - (Eventual) $\varepsilon$ -steadiness

Let  $\varepsilon > 0$ . A sequence  $(a_n)_{n=0}^\infty$  is  $\varepsilon$ -steady if and only if for all natural numbers  $j, k$ ,  $d(a_j, a_k) \leq \varepsilon$ . A sequence  $(a_n)_{n=0}^\infty$  is eventually  $\varepsilon$ -steady if and only if the subsequence  $(a_n)_{n=N}^\infty$  is  $\varepsilon$ -steady for some  $N \in \mathbb{N}$ .

### Problem 1 - Examples and Non-examples of (eventual) $\varepsilon$ -steadiness

We show examples and non-examples of (eventual)  $\varepsilon$ -steady sequences.

**Proposition 1.1.** *The sequence  $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$  is 0.1-steady but not 0.01-steady.*

*Proof.* For the sequence  $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$ , observe that for any positive natural numbers  $j, k$  we have that  $0 \leq 10^{-j}, 10^{-k} \leq 0.1$ . Notice that  $|10^{-j} - 10^{-k}| \leq \max\{10^{-j}, 10^{-k}\} \leq 0.1$ . Hence, the sequence is 0.1-steady.

However, the sequence is not 0.01-steady because  $|a_1 - a_2| = |0.1 - 0.01| = 0.09 \geq 0.01$ .  $\square$

**Proposition 1.2.** *The sequence  $(a_n) = 2^n = 1, 2, 4, 8, \dots$  is not  $\varepsilon$ -steady for any  $\varepsilon > 0$ .*

*Proof.* Suppose for the sake of a contradiction that the sequence  $(a_n) = 2^n = 1, 2, 4, 8, \dots$  is  $\varepsilon$ -steady for some  $\varepsilon_0 > 0$ . Then, it would imply that for all positive numbers  $j, k$ , we have  $|2^j - 2^k| \leq \varepsilon_0$ . Pick  $k = 0$ , so  $|2^j - 1| \leq \varepsilon_0$  implies  $2^j \leq 1 + \varepsilon_0$ . Then, by Proposition 4.4.1, we can find a natural number  $N$  such that  $1 + \varepsilon_0 < N$ . Hence, we have that  $2^j \leq 1 + \varepsilon_0 < N$ , so  $2^j < N$ . If we choose  $j := N$ , we have that  $2^N < N$ , which contradicts the fact that  $2^N \geq N$  for all natural numbers  $N$ .  $\square$

**Proposition 1.3.** *The sequence  $10, 0, 0, 0, \dots$  is not  $\varepsilon$ -steady for any  $\varepsilon < 10$ , but is eventually  $\varepsilon$ -steady for  $\varepsilon > 0$ .*

*Proof.* The sequence  $10, 0, 0, 0, \dots$  is not  $\varepsilon$ -steady for any  $\varepsilon < 10$  because we have that  $|10 - 0| = 10 \not\leq \varepsilon$ . However, the sequence is eventually  $\varepsilon$ -steady if we choose  $N = 1$ . This is because for all  $j, k \geq 1$ , we have  $|a_j - a_k| = 0 < \varepsilon$ .  $\square$

### Definition 2 - Cauchy Sequences

A sequence  $(a_n)_{n=0}^\infty$  is a Cauchy sequence if and only if for all  $\varepsilon > 0$ , the sequence is eventually  $\varepsilon$ -steady. In other words,  $\forall \varepsilon > 0, \exists N \in \mathbb{N}$  such that  $j, k \in \mathbb{N} \implies d(a_j, a_k) \leq \varepsilon$ .

### Definition 3 - Boundedness

Let  $M \geq 0$  be rational. A finite sequence  $a_1, a_2, \dots, a_n$  is bounded by  $M$  if and only if  $|a_i| \leq M$  for all  $1 \leq i \leq n$ . An infinite sequence is bounded by  $M$  if and only if  $|a_i| \leq M$  for all  $i \geq 0$ . Overall, a sequence is said to be bounded if and only if it is bounded by  $M$  for some rational  $M \geq 0$ .

**Proposition 1.4.** *The sequence  $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$  is unbounded.*

*Proof.* Suppose the sequence  $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$  is bounded, so there exists a rational number  $M \geq 0$  such that  $|a_n| = n \leq M$  for all  $n \geq 1$ . By Proposition 4.4.1, there exists a natural number  $N$  such that  $N > M$ . But by our supposition, we also have that  $|a_N| = N \leq M$ , a contradiction!  $\square$

### Problem 2 - Cauchy sequences are bounded

Every Cauchy sequence  $(a_n)_{n=0}^\infty$  is bounded.

*Proof.* Suppose the sequence  $(a_n)_{n=0}^\infty$  is a Cauchy sequence, so we choose  $\varepsilon = 1$ . Then, there exists a natural number  $N$  such that for all  $j, k \geq N$ , we have  $d(a_j, a_k) = |a_j - a_k| \leq 1$ . Then, by the reverse triangle inequality,  $||a_j| - |a_k|| \leq |a_j - a_k| \leq 1$ . By simplifying, we get that  $|a_j| \leq 1 + |a_k|$ . Pick  $k := N$  so that for any  $j \geq N$ , we have  $|a_j| \leq 1 + |a_N|$ . Therefore, for the subsequence of  $(a_n)_{n=N}^\infty$ , it is bounded by  $1 + |a_N|$ . Then, for the finite subsequence  $(a_n)_{n=0}^N$ , there exists  $M$  such that  $M$  bounds the finite subsequence. Therefore, the whole sequence is bounded by  $\max\{1 + |a_N|, M\}$ .  $\square$

## B: Equivalent Cauchy Sequences

### Definition 4 - Equivalent Sequences

Two sequences  $(a_n)_{n=0}^\infty$  and  $(b_n)_{n=0}^\infty$  are equivalent if and only if for every  $\varepsilon > 0$ , the sequences are eventually  $\varepsilon$ -close. In other words,  $\exists N \in \mathbb{N}$  such that  $\forall n \geq N$ ,  $|a_n - b_n| < \varepsilon$ .

### Problem 1 - Equivalence of Cauchy Sequences

If  $(a_n)_{n=0}^\infty$  and  $(b_n)_{n=0}^\infty$  are equivalent sequences of rationals, then  $(a_n)_{n=0}^\infty$  is a Cauchy sequence if and only if  $(b_n)_{n=0}^\infty$  is a Cauchy sequence.

*Proof.* Suppose  $(a_n)_{n=0}^\infty$  and  $(b_n)_{n=0}^\infty$  are equivalent sequences of rationals, and suppose also  $(a_n)_{n=0}^\infty$  is a Cauchy Sequence. We now show that  $(b_n)_{n=0}^\infty$  is also Cauchy. Let  $\varepsilon > 0$  be given.

By the equivalence of the two sequences, there exists a natural number  $N$  such that if  $n \geq N_1$ , then  $d(b_n, a_n) \leq \frac{\varepsilon}{4}$ . Moreover, we also have by  $(a_n)_{n=0}^\infty$  being Cauchy that there exists a natural number  $M$  such that if  $j, k \geq M$ , then  $d(a_j, a_k) \leq \frac{\varepsilon}{2}$ . We choose  $L = \max\{N, M\}$  so that we have the useful properties from equivalence and Cauchy sequences. Then, by one of the properties of the distance function (specifically of “If  $d(x, y) \leq \varepsilon$  and  $d(z, w) \leq \delta$ , then  $d(x \pm z, y \pm w) \leq \varepsilon + \delta$ ”), we have that

$$\begin{aligned}
d(b_j, a_j) \leq \frac{\varepsilon}{4} \text{ and } d(b_k, a_k) \leq \frac{\varepsilon}{4} &\implies d(b_j - b_k, a_j - a_k) \leq \frac{\varepsilon}{2} && \text{by properties of the distance function} \\
&\implies |(b_j - b_k) - (a_j - a_k)| \leq \frac{\varepsilon}{2} && \text{by definition of distance} \\
&\implies ||b_j - b_k| - |a_j - a_k|| \leq \frac{\varepsilon}{2} && \text{by the reverse triangle inequality} \\
&\implies |b_j - b_k| - |a_j - a_k| \leq \frac{\varepsilon}{2} && \text{because we only care about being } \leq \frac{\varepsilon}{2} \\
&\implies |b_j - b_k| - \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2} && \text{by } (a_n)_{n=0}^\infty \text{ being a Cauchy sequence} \\
&\implies d(b_j, b_k) \leq \varepsilon && \text{by rearrangement.}
\end{aligned}$$

For the converse, just switch the two sequences.  $\square$

### Problem 2 - Boundedness of Equivalent Cauchy Sequences

Let  $\varepsilon > 0$ . Show that if  $(a_n)_{n=0}^\infty$  and  $(b_n)_{n=0}^\infty$  are eventually  $\varepsilon$ -close, then  $(a_n)_{n=0}^\infty$  is bounded if and only if  $(b_n)_{n=0}^\infty$  is bounded.

*Proof.* Let  $\varepsilon > 0$  be given. Suppose the sequences  $(a_n)_{n=0}^\infty$  and  $(b_n)_{n=0}^\infty$  are eventually  $\varepsilon$ -close, so they are equivalent sequences. Suppose also that  $(a_n)_{n=0}^\infty$  is bounded, so there exists a non-negative rational  $M$  such that  $|a_i| \leq M$  for all  $i \geq 0$ . We wish to show that  $(b_n)_{n=0}^\infty$  is also bounded.

Because the two sequences are eventually  $\varepsilon$ -close, without loss of generality, we arbitrarily picked  $\varepsilon := 1$ . We then choose  $N$  such that for all  $n \geq N$ , we have that  $|b_n - a_n| \leq 1$ . Then, by the reverse triangle inequality, we have that  $||b_n| - |a_n|| \leq |b_n - a_n| \leq 1$ . By transitivity, we get  $||b_n| - |a_n|| \leq 1$ . By the properties of the absolute value, we have that  $-1 \leq |b_n| - |a_n| \leq 1$ . Focusing on the middle and right hand side of the inequalities, we get  $|b_n| - |a_n| \leq 1$ . Adding  $|a_n|$  to both sides gives  $|b_n| \leq 1 + |a_n|$ . Because of  $(a_n)_{n=0}^\infty$  being bounded, we have that  $|a_n| \leq M$ . By substitution, we get  $|b_n| \leq 1 + M$ . Therefore, for the tail of the infinite subsequence  $(b_n)_{n=N}^\infty$ , it is bounded by  $M + 1$ . For the head of the finite subsequence, it is bounded by some  $L := \max\{|b_0|, |b_1|, \dots, |b_{N-1}|\}$ . Hence, the overall subsequence  $(b_n)_{n=0}^\infty$  is bounded by  $\max\{1 + M, L\}$ .

For the converse, just switch the sequences being treated. The logic follows because the distance function is symmetric.  $\square$

## C: The Construction of Real Numbers

### Definition 5 - Real Numbers ( $\mathbb{R}$ )

A real number is defined to be an object of the form  $\text{LIM}_{n \rightarrow \infty} a_n$ , where  $(a_n)_{n=0}^{\infty}$  is a Cauchy sequence of rational numbers. Two real numbers  $\text{LIM}_{n \rightarrow \infty} a_n$  and  $\text{LIM}_{n \rightarrow \infty} b_n$  are said to be equal if and only if  $(a_n)_{n=0}^{\infty}$  and  $(b_n)_{n=0}^{\infty}$  are equivalent Cauchy sequences.